

Instructions: This exam contains 6 questions, each of 12 points. Each question has two parts, each part is worth 6 points. Maximum possible points is 60; there are 12 bonus points. Attempt as many as you can. Time allotted is 2 hours.

Question 1. Let $(S_n)_{n=0}^\infty$ be a simple symmetric random walk starting from 0.

1. What is the probability that $S_{20} = S_{30} = 6$?
2. What is the probability that $S_{20} = 6$ and $S_k \leq -2$ for some $k \in \{1, 2, \dots, 19\}$.

Question 2. Let $(S_n)_{n=0}^\infty$ be a simple asymmetric random walk starting from 0. Suppose each step is either 1 with probability 0.75 or -1 with probability 0.25.

1. What is the probability that $S_{20} = S_{30} = 6$?
2. What is the probability that the walk hits 6 and does not hit -10 before it hits 6?

Question 3. Let $X_1, X_2, \dots, X_n$ be i.i.d. with density function $f(x) = \frac{1}{x^2}$ for $x > 1$.

1. Find the distribution function and the density function of $X_{(1)}$.
2. Find the distribution function and the density function of $X_{(n)}$.

Question 4. Let $U_1, U_2, \dots$ be i.i.d. Uniform $[0, 1]$ random variables. For each $n \geq 1$, let A_n be the event that $U_n \leq 1/n^2$, and let B_n be the event that $U_1 \leq 1/n$.

1. What is the probability of the event $\limsup A_n$?
2. What is the probability of the event $\limsup B_n$?

Question 5. Suppose $(X_n)_{n=1}^\infty$ is a sequence of independent random variables such that $X_n = 0$ with probability p_n and $X_n = a_n$ with probability $1 - p_n$. Here $(a_n)_{n=1}^\infty$ is a sequence in $(0, \infty)$ and $(p_n)_{n=1}^\infty$ is a sequence in $(0, 1)$.

1. Find an example of $(a_n)_{n=1}^\infty$ and $(p_n)_{n=1}^\infty$ such that $(X_n)_{n=1}^\infty$ converges to 0 in probability but not almost surely.
2. Find an example of $(a_n)_{n=1}^\infty$ and $(p_n)_{n=1}^\infty$ such that $(X_n)_{n=1}^\infty$ converges to 0 in probability but not in L^1 .

Question 6. Suppose $(X, Y) \sim \mathcal{N}(1, 1, 4, 4, 0.5)$

1. What is the joint distribution of $X + Y$ and $X - Y$?
2. What is the conditional distribution of X given Y ?